

4.6 Solution: When only quadrupole moment is nonzero, the energy becomes

$$W = -\frac{e}{6} \sum_{ij} Q_{ij} \left(\frac{\partial E_j}{\partial x_i} \right)_0.$$

If we have $Q_{zz} = Q$ and $Q_{xx} = Q_{yy} = -Q/2$ by tracelessness of the quadrupole moment, we can write the energy as

$$\begin{aligned} W &= -\frac{e}{6} \left[Q_{xx} \left(\frac{\partial E_x}{\partial x} \right)_0 + Q_{yy} \left(\frac{\partial E_y}{\partial y} \right)_0 + Q_{zz} \left(\frac{\partial E_z}{\partial z} \right)_0 \right] \\ &= -\frac{e}{6} Q \left[-\frac{1}{2} \left(\frac{\partial E_x}{\partial x} \right)_0 - \frac{1}{2} \left(\frac{\partial E_y}{\partial y} \right)_0 + \left(\frac{\partial E_z}{\partial z} \right)_0 \right]. \end{aligned}$$

By $\nabla \cdot \mathbf{E} = 0$, we know

$$\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} = -\frac{\partial E_z}{\partial z},$$

then

$$W = -\frac{e}{6} Q \left[\frac{1}{2} + 1 \right] \left(\frac{\partial E_z}{\partial z} \right)_0 = -\frac{e}{4} Q \left(\frac{\partial E_z}{\partial z} \right)_0.$$